

(1) Murphy 5.40

- (2) Murphy 5.41 The figure included in the problem statement shows c_p in mmol/L plotted on the horizontal axis and q_p in mmol/g plotted on the vertical axis. The data are given at the right. Fit these data to two adsorption models: (1) the Langmuir isotherm:

$$q_p = q_\infty \frac{K_L c_p}{1 + K_L c_p}$$

where q_∞ and K_L are adjustable parameters and (2) the Freundlich isotherm:

$$q_p = k_F c_p^{1/n}$$

where k_F and n are another set of adjustable parameters. Try fitting the data to both models. Which is better? Use that model to solve the problem.

c_p (mmol/L)	q_p (mmol/g)
0.02	0.79
0.11	1.25
0.36	1.65
0.43	1.86
0.60	1.88
0.89	2.11
0.98	2.18
1.81	2.56
2.30	2.72
3.06	2.79
3.59	2.90
4.05	3.01
4.71	2.99
5.27	3.17
6.05	3.21
6.69	3.30

- (3) Air is heated from 25°C to 150°C prior to entering a combustion furnace. The change in specific enthalpy associated with this transition is 3640 J/mol. The flow rate of air at the heater outlet is 1.25 m³/min and the air pressure at this point is 122 kPa absolute.

- (a) Calculate the heat requirement in kW, assuming ideal gas behavior and that the kinetic and potential energy changes from the heater inlet to the outlet are negligible.
- (b) Would the value of $\Delta \hat{E}_k$ [which was neglected in part (a)] be positive or negative, or would you need more information to answer? If you need more information, what would that information be?

- (4) Murphy 6.26 (For parts (a) through (e), which are all changes that are described by classical thermodynamics, the change in energy is the sum of the change in internal energy plus the change in kinetic energy plus the change in potential energy. Parts (c), (d), and (e) all are asking for the internal energy change for the change in state specified. This is because you are considering a specified amount of material, namely, one gram mole of H₂O, which is a closed system.)

0 → 100 km/h

a) $v = 100 \text{ km/h} = 27.78 \text{ m/s}$

$m = 18 \text{ g} = 0.018 \text{ kg}$
(1 g mol H₂O)

$\Delta KE = \frac{1}{2} m v^2 = \frac{1}{2} (0.018) (27.78)^2$
⇒ $\Delta KE = 6.94 \text{ J}$

b) Rising 100 m

$\Delta PE = mgz = (0.018) (9.81) (100)$
⇒ $\Delta PE = 17.7 \text{ J}$

c) Pressure increase @ 100°C (0.1 → 1 bar)
T const for liquid water.

⇒ $\Delta U = 0 \text{ J}$ (independent of pressure)

d) Heating 0 → 100°C @ 5 bar.

$\Delta U = n C_v \Delta T = (1) (75 \text{ J/mol} \cdot \text{K}) (100 \text{ K})$
⇒ $\Delta U = 7500 \text{ J}$

e) Dissociation to H₂ and O₂ @ 25°C.



• $\Delta H^\circ = + 285.8 \text{ kJ/mol} = 285800 \text{ J/mol}$

• $\Delta U = \Delta H - \Delta(PV)$

• $\Delta(PV) = \Delta n RT = [(1 + 0.5) - 1] (8.314) (298) = 1238 \text{ J}$

⇒ $\Delta U = 285800 - 1238 = 284600 \text{ J}$
 $\approx 2.85 \times 10^5 \text{ J}$

23/25

3) a)

$$P = 122 \text{ kPa} = 122000 \text{ Pa}.$$

$$\dot{V} = 1.25 \text{ m}^3/\text{min} = \frac{1.25}{60} = 0.02083 \text{ m}^3/\text{s}.$$

$$T = 150^\circ\text{C} = 423.15 \text{ K}.$$

$$R = 8.314 \text{ J/mol}\cdot\text{K}.$$

$$\Rightarrow \dot{n} = \frac{P\dot{V}}{RT} = \frac{(122000)(0.02083)}{(8.314)(423.15)} = 0.722 \text{ mol/s}.$$

$$\Delta H = 3640 \text{ J/mol} \Rightarrow \dot{Q} = \dot{n} \Delta H = (0.722)(3640) = 2629.6 \text{ J/s}.$$

$$\Rightarrow \dot{Q} = 2.63 \text{ kW}.$$

$$b) \Delta \dot{E}_k = \dot{m} \left(\frac{V_2^2 - V_1^2}{2} \right)$$

• If $V_2 > V_1 \rightarrow$ the air speed up $\rightarrow$ positive. $\left\{ \rightarrow \text{More info is required.} \right.$

• If $V_2 < V_1 \rightarrow$ the air slow down $\rightarrow$ negative. $\left\{ \rightarrow \text{he cannot determine the sign.} \right.$

So, velocity is needed @ both inlet and outlet.

pipe diameters/cross-sectional areas @ inlet and outlet.

volumetric flow rates @ both inlet and outlet.

1) 5.40

Feed: 25 wt% acetic acid (A) $\rightarrow$ 25 kg (A)

75 wt% water (W) $\rightarrow$ 75 kg (W)

$$F = 100 \text{ kg}.$$

Solvent: pure MIBK (S) $\rightarrow$ 0 kg

Mixer-settler

Requirement: raffinate (water-rich phase) $\leq 6 \text{ wt\% (A)}$.

From the diagram:

	Raffinate (6 wt% A)	Extract
A	$\approx 6 \text{ wt\%}$	$\approx 20 \text{ wt\%}$
MIBK	$\approx 2 \text{ wt\%}$	$\approx 75 \text{ wt\%}$
W	$\approx 92 \text{ wt\%}$	$\approx 5 \text{ wt\%}$

Mixing point: M (20% A, 65% MIBK)

Lever rule on F-S line:

$$\frac{S}{F} = \frac{FM}{MS} = \frac{45}{35} = 1.29 \Rightarrow S = 100 \times 1.29 = 129 \text{ kg MIBK}$$

$$\Rightarrow m_{\text{total}} = F + S = 100 + 129 = 229 \text{ kg}.$$

Lever rule on tie line:

$$\frac{E}{R} = \frac{MR}{ME} = \frac{25}{40} = 0.625 \Rightarrow E = 0.625R \Rightarrow E = 88 \text{ kg}.$$

$$m = E + R = 0.625R + R = 229 \Rightarrow R = 141 \text{ kg}$$

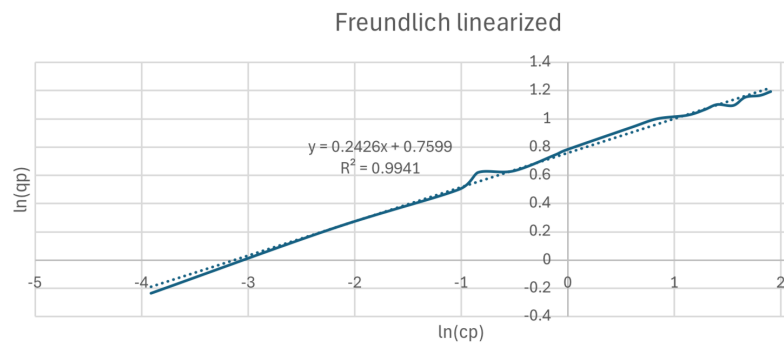
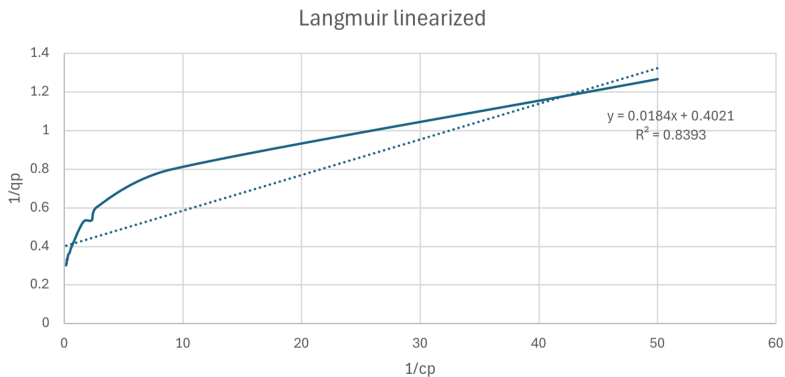


	A	B	C	D	E	F
1	cp(mmol/L)	qp(mmol/g)	1/cp	1/qp	ln(cp)	ln(qp)
2	0.02	0.79	50	1.265822785	-3.912023005	-0.235722334
3	0.11	1.25	9.090909091	0.8	-2.207274913	0.223143551
4	0.36	1.65	2.777777778	0.606060606	-1.021651248	0.500775288
5	0.43	1.86	2.325581395	0.537634409	-0.84397007	0.620576488
6	0.6	1.88	1.666666667	0.531914894	-0.510825624	0.631271777
7	0.89	2.11	1.123595506	0.473933649	-0.116533816	0.746687947
8	0.98	2.18	1.020408163	0.458715596	-0.020202707	0.779324877
9	1.81	2.56	0.552486188	0.390625	0.593326845	0.940007258
10	2.3	2.72	0.434782609	0.367647059	0.832909123	1.00063188
11	3.06	2.79	0.326797386	0.358422939	1.118414916	1.026041596
12	3.59	2.9	0.278551532	0.344827586	1.278152203	1.064710737
13	4.05	3.01	0.24691358	0.332225914	1.398716881	1.101940079
14	4.71	2.99	0.212314225	0.334448161	1.549687908	1.095273387
15	5.27	3.17	0.189753321	0.315457413	1.662030363	1.153731588
16	6.05	3.21	0.165289256	0.31152648	1.800058272	1.166270937
17	6.69	3.3	0.149476831	0.303030303	1.900613874	1.193922468
18						

Langmuir Results

slope m	0.018416307	slope 1/n	0.242642968
intercept b	0.402050886	intercept ln(kF)	0.7599369
q_infinity	2.487247346	n	4.121281606
KL	21.831244	kF	2.1381413
R^2	0.839298603	R^2	0.994133266

Freundlich Results



	L	M	N	O
1		Part 2 - Equilibrium Calculation		
2		Molecular weight of phenol (g/mol)	94.11	
3		Initial phenol mass (g)	1	
4		Initial phenol amount (mol)	0.010625863	
5		Initial phenol amount (mmol)	10.62586335	
6		Solution volume (L)	1	
7		Activated carbon mass (g)	5	
8		kF	2.1381413	
9		n	4.121281606	
10		Ce guess (mmol/L)	0.727933838	equilibrium concentration in solution
11		qe (mmol/g)	1.979584176	equilibrium loading on carbon, qe
12		Phenol adsorbed (mmol)	9.89792088	mmol phenol adsorbed onto 5 g carbon
13		Phenol remaining in solution (mmol)	0.727933838	mmol phenol left in 1 L solution
14		Total phenol at equilibrium (mmol)	10.62585472	total mmol after equilibrium
15		Difference from initial (mmol)	-8.63392E-06	error in mass balance
16		Fraction adsorbed	0.931493334	
17		Percent adsorbed	93.1493334	
18				
19			ANS: 93.15	
20				
21				
22				

I used Goal Seek
Set cell: N15
To value: 0
By changing cell: N10
Then N10 to N17 changed

93.15